

Comprehensive EDA Report: Customer & Transaction Analysis

1. Executive Summary

This report presents an exhaustive Exploratory Data Analysis (EDA) of a high-volume retail dataset containing ~31.7M transactions, 1.37M customers, and 105k articles. The analysis reveals a “Fast Fashion” dynamic characterized by extreme matrix sparsity (99.98%), significant customer churn (45.5% inactive > 6 months), and high SKU turnover (short article shelf life). Despite these challenges, the data exhibits perfect referential integrity and strong demographic segmentation, providing a solid foundation for advanced recommendation systems and customer lifetime value (CLV) modeling.

2. Data Dictionary & Structural Metadata

The dataset consists of three core tables processed into Parquet format using `polars` for optimal memory efficiency and I/O speed.

2.1. Articles (`articles.parquet`)

- Purpose: Master catalog of all products with rich categorical metadata.
- Scale: 105,542 rows | 25 columns | Size: ~36.38 MB (Parquet)

Feature	Type	Null %	Range / Uniques	Sample	Purpose	
<code>article_id</code>	Int64	0.00%	[108775015, 959461001]	108775015	Primary Key	
<code>product_code</code>	Int64	0.00%	[108775, 959461]	108775	High-level identifier	SKU
<code>prod_name</code>	String	0.00%	45,875 unique	Strap top	Product Name	
<code>product_type_name</code>	String	0.00%	131 unique	Vest top	Granular category	
<code>product_group_name</code>	String	0.00%	19 unique	Garment body	Upper	High-level category
<code>graphical_appearance_name</code>	String	0.00%	30 unique	Solid	Pattern/Texture	
<code>colour_group_name</code>	String	0.00%	50 unique	Black	Primary color	
<code>perceived_colour_value_name</code>	String	0.00%	8 unique	Dark	Color tone	
<code>department_name</code>	String	0.00%	250 unique	Jersey Basic	Internal department	

Feature	Type	Null %	Range / Uniques	Sample	Purpose
index_name	String	0.00%	10 unique	Ladieswear	Macro department
section_name	String	0.00%	56 unique	Womens Every-day Basics	Store section
garment_group_name	String	0.00%	21 unique	Jersey Basic	Manufacturing group
detail_desc	String	0.39%	43,405 unique	Jersey top with...	Product description

2.2. Customers (`customers.parquet`)

- Purpose: Customer demographic and loyalty program metadata.
- Scale: 1,371,980 rows | 7 columns | Size: ~214.99 MB (Parquet)

Feature	Type	Null %	Range / Uniques	Sample	Purpose
customer_id	String	0.00%	1,371,980 unique	00000dbacae5...	Hexadecimal Primary Key
FN	Float64	65.24%	[1.0, 1.0]	1.0	Fashion News flag (Null = No)
Active	Float64	66.15%	[1.0, 1.0]	1.0	Active profile flag (Null = No)
club_member_status	String	0.44%	4 unique	ACTIVE	Membership lifecycle status
fashion_news_frequency	String	1.17%	5 unique	NONE	Comm frequency
age	Int64	1.16%	[16, 99]	49	Customer age
postal_code	String	0.00%	352,899 unique	52043ee2162c...	Hashed location identifier

2.3. Transactions (`transactions_train.parquet`)

- Purpose: The core interaction log between customers and articles.
- Scale: 31,788,324 rows | 5 columns | Size: ~2.97 GB (Parquet)

Feature	Type	Null %	Range / Uniques	Sample	Purpose
t_dat	String	0.00%	734 unique days	2018-09-20	Transaction Date

Feature	Type	Null %	Range / Uniques	Sample	Purpose
customer_id	String	0.00%	1,362,281 unique	000058a12d5b...	Foreign Key (Customers)
article_id	Int64	0.00%	[108775015, 956217002]	663713001	Foreign Key (Articles)
price	Float64	0.00%	[1.69e-05, 0.591]	0.0508	Normalized Price
sales_channel_id	Int64	0.00%	[1, 2]	2	Channel ID (1: Online, 2: Store)

3. Data Integrity & Foundational Quality

3.1. Missingness & Completeness

- **Referential Integrity:** 100%. Every `customer_id` and `article_id` present in `transactions_train.parquet` successfully maps to a record in their respective metadata tables.
- **Null Values:**
 - The `transactions` table is perfectly complete (0% nulls).
 - In `customers`, `FN` and `Active` flags have ~65% nulls. These are binary flags where `Null` should be safely imputed as `0` or `False`.
 - `age` has a 1.16% missing rate, requiring simple median imputation or age-bucket prediction for downstream modeling.

3.2. Redundancy & Duplication

- **Exact Duplicates:** The `transactions` table contains 5,518,813 duplicate rows (17.36%).
- **Interpretation:** Given the absence of a `quantity` column, an exact match on `(t_dat, customer_id, article_id, price, sales_channel_id)` represents a customer buying multiple units of the exact same SKU in a single transaction event.
- **Technical Note:** Identified using Polars' high-performance duplicate detection: `df.filter(pl.struct(df.columns).is_duplicated())`. These rows must be grouped and aggregated into a `quantity` feature prior to model training to prevent target leakage or skewed engagement metrics.

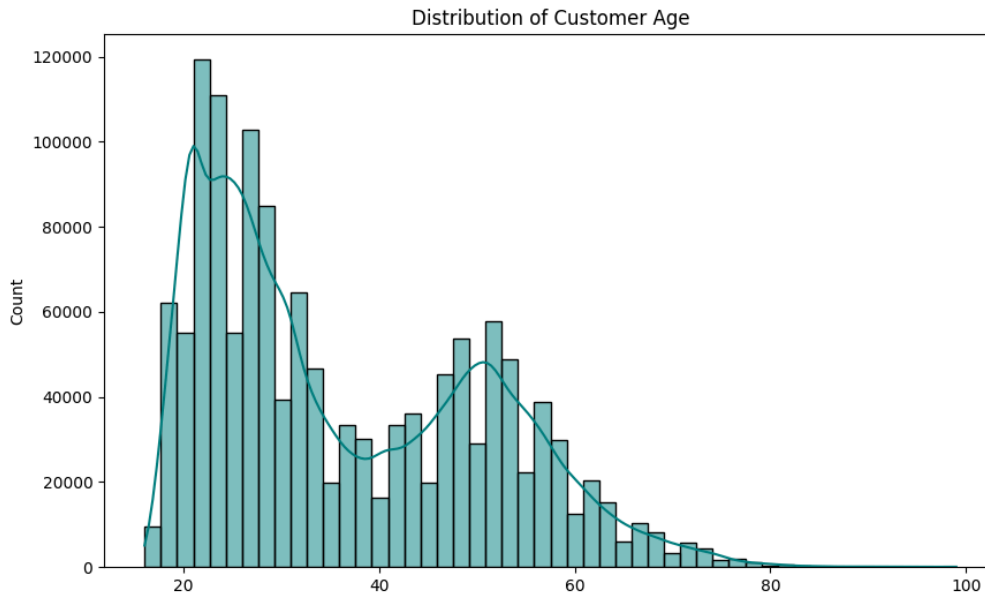
4. Univariate & Demographic Analysis

4.1. The Bimodal Customer Base

Age distribution analysis reveals a heavily bimodal demographic structure.

- **Primary Peak (Youth):** High concentration around age 25.
- **Sec-**

ondary Peak (Mature): Significant bump around age 50. - Visual Evidence:



4.2. Category & Sales Channel Dominance

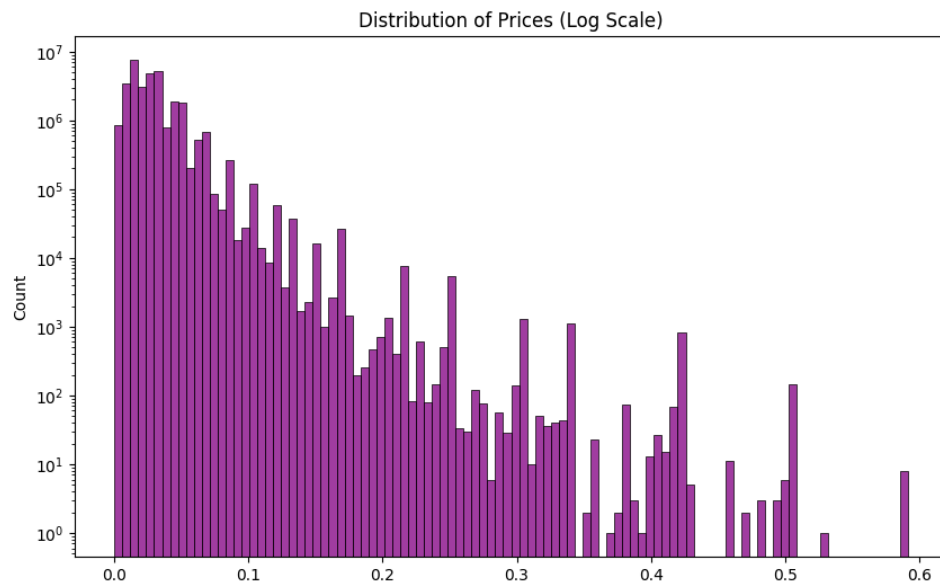
- Sales Channels: channel 2 is the dominant purchasing avenue, accounting for ~71% of all transactions, while channel 1 captures the remaining 29%.
- Index Dominance: Ladieswear is the absolute market leader. Divided and Menswear follow but at significantly lower volumes.

4.3. Price Distribution & Normality

- Distribution Shape: Transaction prices follow a strict log-normal distribution.
- Ranges: The vast majority of items sit in the normalized $[0.01, 0.1]$ band. The overall mean is ~ 0.027 .

- Visual

Evidence:



5. Temporal & Seasonal Trends

5.1. Macro Seasonality & Volume Spikes

- Cyclical Trends: Transaction volumes exhibit strong seasonality, with significant spikes during Spring/Summer (April-July) and deep Winter (Black Friday through December).
- Weekly Patterns: Saturday represents the peak shopping day, closely followed by Wednesday.
- Visual Evidence:

►

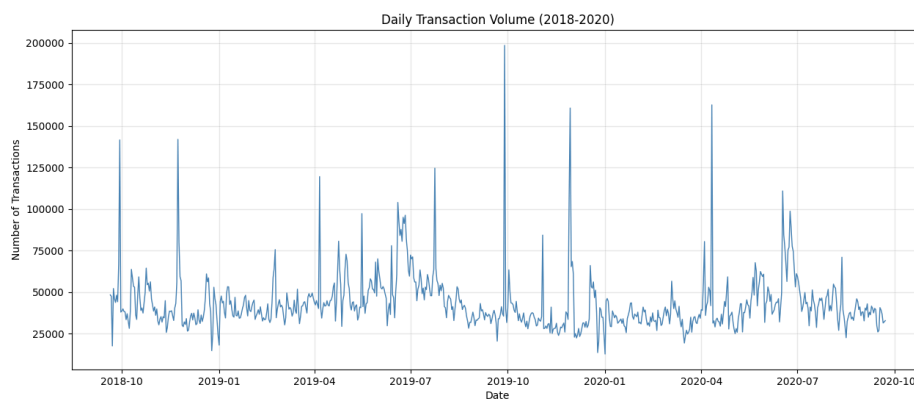


Figure 1: Daily Volume

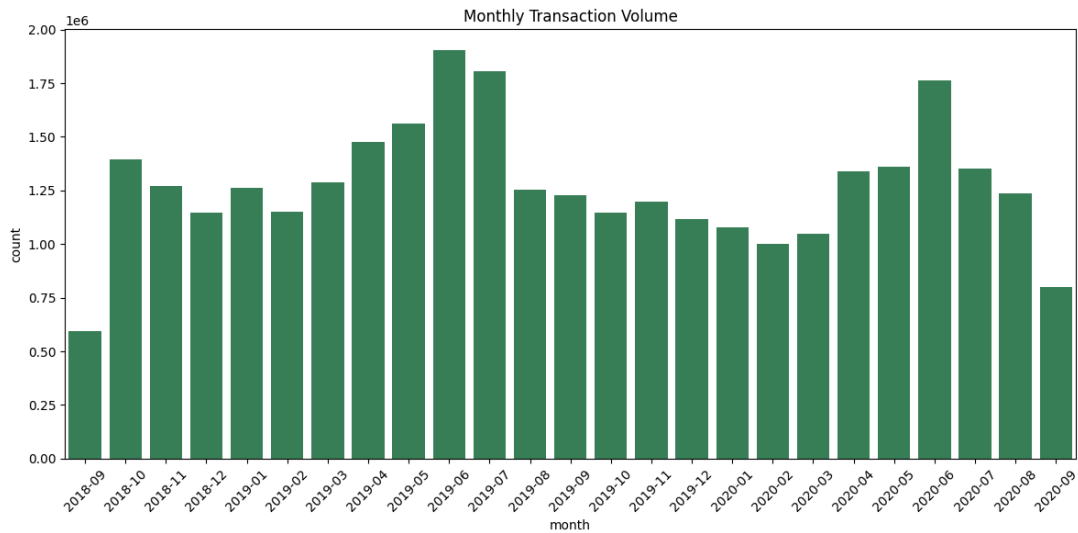
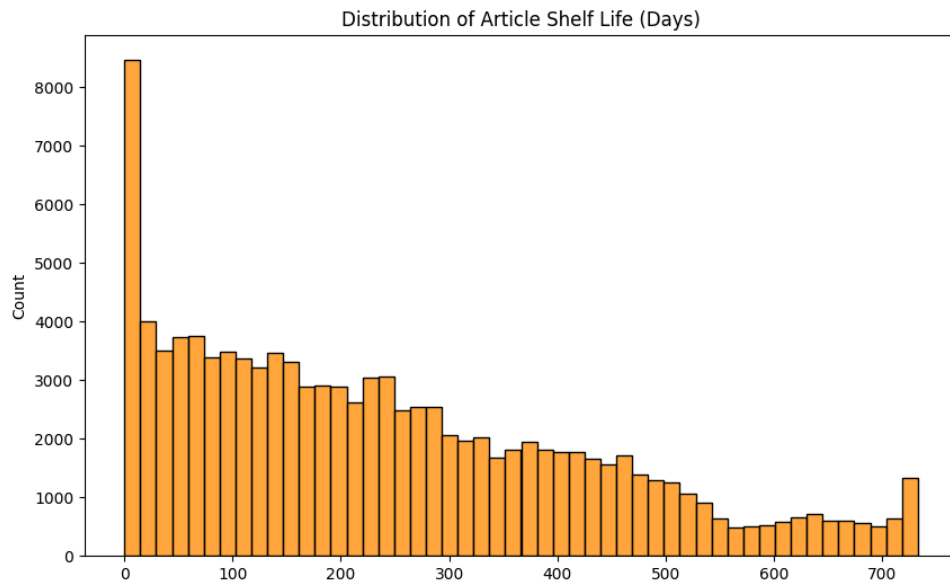


Figure 2: Monthly Volume

5.2. Article Turnover (Shelf Life)

- Fast Fashion Lifecycle: The calculated “Shelf Life” (delta between first and last recorded sale per `article_id`) highlights that a significant portion of the catalog is active for fewer than 90 days.
- Cold Start Implication: This rapid turnover mandates that recommendation models heavily weight “freshness” and rely on content-based metadata, as collaborative filtering will continuously struggle with new, interaction-poor items.
- Visual Evidence:



6. Anomalies, Extremes, & Geographical Concentration

6.1. The “Super-Code” Anomaly

- Finding: Postal code `2c29ae653a9282cce4151bd87643c9...` is responsible for ~626,000 transactions, whereas the second-highest code accounts for only ~5,800.
- Interpretation: This extreme outlier is almost certainly a default placeholder for “Unknown/Null” locations or a massive aggregator account. It should be explicitly flagged or masked in regional models.

6.2. Resellers & Power Users

- Volume Outliers: The top customer in the dataset purchased 18,950 items over the two-year period (averaging ~25 items/day). The top 10 customers all exceed 8,000 items.
- Interpretation: These are clear indicators of reseller behavior or wholesale accounts. Their interactions will skew standard collaborative filtering approaches.

6.3. Extreme Price Volatility

- Swings: While the mean price is ~0.027, the absolute maximum reaches ~0.59. More critically, specific articles exhibit extreme internal price volatility. For instance, article `715052001` has a max/min price ratio of 1499x.
- Interpretation: Such dramatic swings typically indicate either deep “penny” clearance events at the end of a season or the recycling of old `article_ids` for new products in legacy systems.

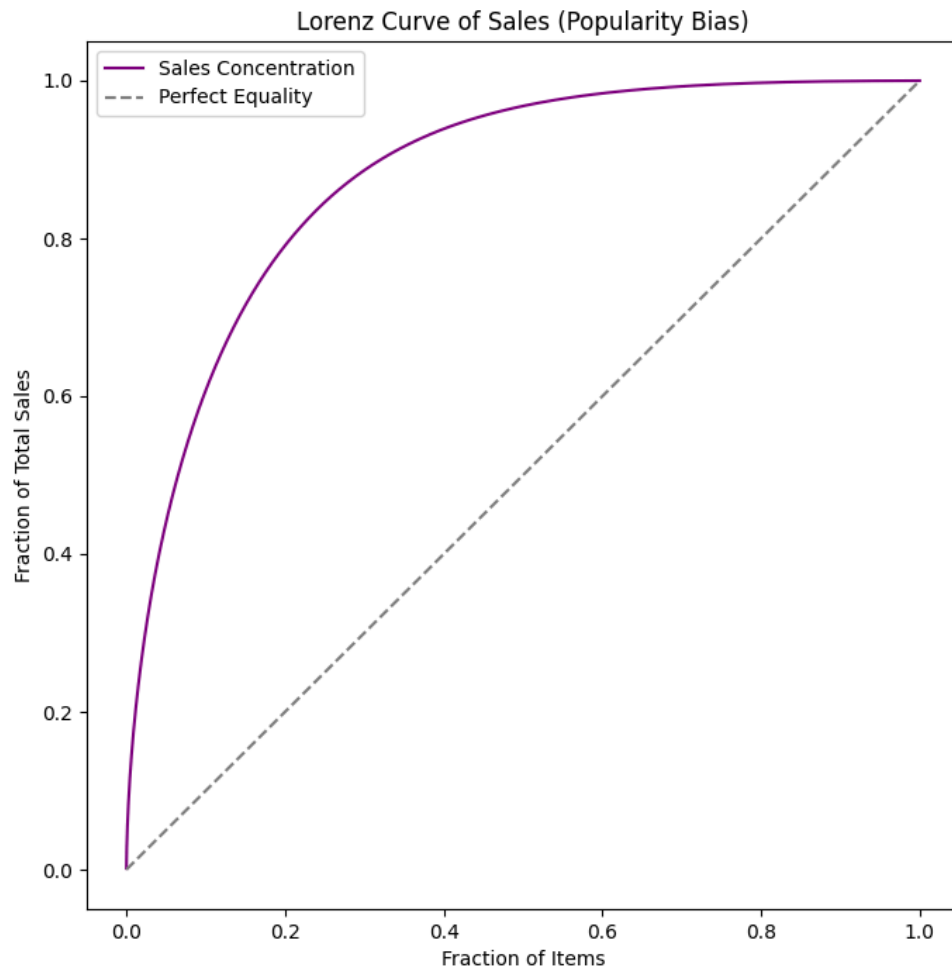
7. Advanced Analytics: RecSys & Strategic Retention

7.1. Matrix Sparsity & Popularity Bias

- Sparsity: The user-item interaction matrix exhibits 99.978% sparsity.
- Lorenz Concentration: Sales are heavily concentrated in a small percentage of “blockbuster” items, demonstrating a steep Gini coefficient/Lorenz curve.
- RecSys Design: Standard Matrix Factorization will fail here. Hybrid models leveraging dense item metadata and sequential transformers (e.g., SASRec) are strictly required.

- Visual

Evidence:

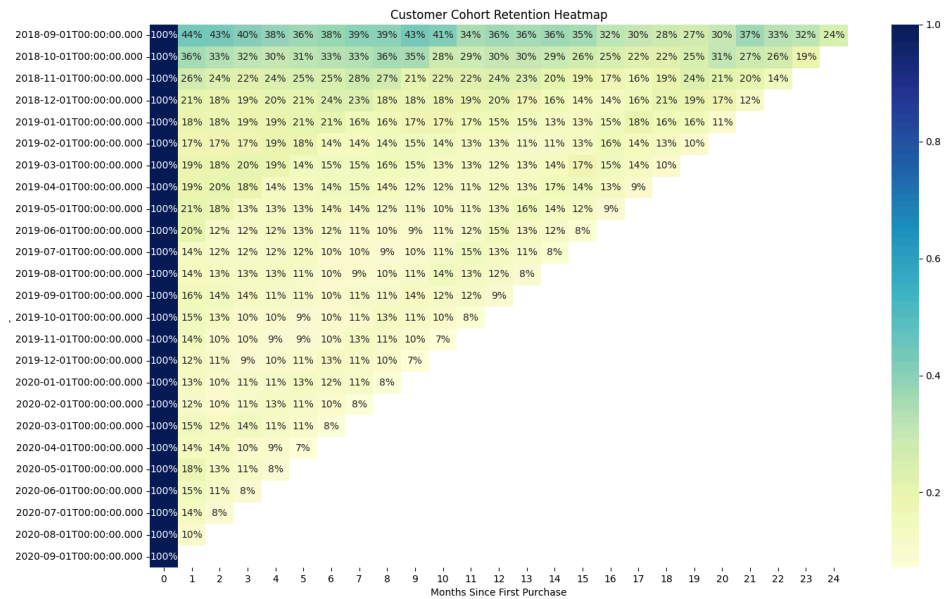


7.2. Cohort Retention & Churn Proxy

- Decay Curve: Cohort retention analysis reveals a massive drop-off after the initial month of purchase. Long-term (2-year) retention hovers below 5%.
- Churn Definition: Defining churn as >180 days inactive, ~45.5% of the customer base is currently "At-Risk". Defining it as >365 days, 27% are fully churned.

- Visual

Evidence:



7.3. Sequential Category Transitions

- The “Ladieswear” Gravity Well: Markov transition analysis of sequential purchases shows that Ladieswear possesses a ~75% stickiness rate (a customer buying Ladieswear has a 75% probability that their next purchase is also Ladieswear).
- Lifecycle Progression: There is a strong transition pipeline from Divided (Youth) to Ladieswear (Mature), mapping the demographic aging of the customer base.
- Technical Note: Sequential next-purchase tracking was achieved via window functions: `df.with_columns(pl.col("index_name").shift(-1).over("customer_id"))`.

- Visual

Evidence:

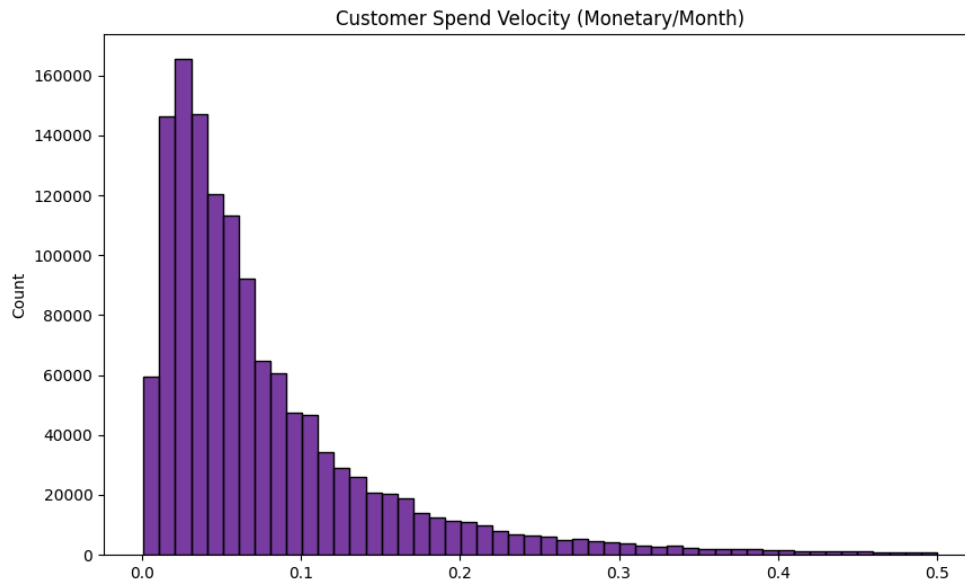


7.4. Customer Spend Velocity

- Metric: By calculating $\text{Total Spend} / \text{Active Months Tenure}$, we isolate high-velocity “Rising Stars” from historical whales who have gone dormant.
- Distribution: While the median velocity is low (<0.1 per month), the long tail extends significantly, identifying a crucial VIP segment for immediate marketing intervention.

- Visual

Evidence:



8. Modeling Roadmap & Feature Engineering Guide

Based on the exhaustive structural and behavioral findings, the following preprocessing and feature engineering steps are strictly recommended prior to model development:

1. Duplicate Aggregation: Group `transactions_train` by `[t_dat, customer_id, article_id, price, sales_channel_id]` and compute the row count as a new `quantity` feature. Drop the exact duplicates.
2. Geographical Masking: Overwrite the “Super-Code” (`2c29ae65...`) in the `customers` table with a standard `Unknown` label to prevent spatial models from collapsing into a single dense node.
3. Null Imputation:
 - `customers.FN` and `customers.Active`: Fill nulls with `0.0`.
 - `customers.age`: Impute with the median (`49`) or build a simple predictive regressor using purchase category affinity (e.g., users who buy `Divided` are highly likely to be in their 20s).
4. RecSys Architecture: Do not attempt standard Collaborative Filtering (CF). The 99.98% sparsity and 90-day SKU shelf life demand a Content-Based Hybrid or Sequential Target model (like SASRec) that embeds article metadata (`index_name`, `garment_group_name`, `detail_desc` N-grams) alongside historical sequences.
5. Retention Target Definition: Use the 180-day inactivity threshold to define binary churn targets for predictive classification models aimed at the “At-Risk” segment.